Upload Dataset

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adult.csv
3.9MB

IncomeScope AI – Income Prediction Dashboard

Interactive analysis, visualization, model evaluation, prediction, and chatbot for the Adult Income dataset.

Dataset uploaded successfully!

Dataset Preview

	age	workclass	fnlwgt	education	education_num	marital_s
0	90	Private	77053	HS-grad	9	Widowec
1	82	Private	132870	HS-grad	9	Widowec
2	66	Private	186061	Some-college	10	Widowec
3	54	Private	140359	7th-8th	4	Divorced
4	41	Private	264663	Some-college	10	Separate
5	34	Private	216864	HS-grad	9	Divorced
6	38	Private	150601	10th	6	Separate
7	74	State-gov	88638	Doctorate	16	Never-m
8	68	Federal-gov	422013	HS-grad	9	Divorced
9	41	Private	70037	Some-college	10	Never-m

Rows
32561

Columns
15

Missing Values
583

Interactive Data Explorer

Choose columns to display

age × workclass × fnlwgt × education × education_num ×

	age	workclass	fnlwgt	education	education_num
0	90	Private	77053	HS-grad	9
1	82	Private	132870	HS-grad	9
2	66	Private	186061	Some-college	10
3	54	Private	140359	7th-8th	4
4	41	Private	264663	Some-college	10
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6	38	Private	150601	10th	6
7	74	State-gov	88638	Doctorate	16
8	68	Federal-gov	422013	HS-grad	9
9	41	Private	70037	Some-college	10

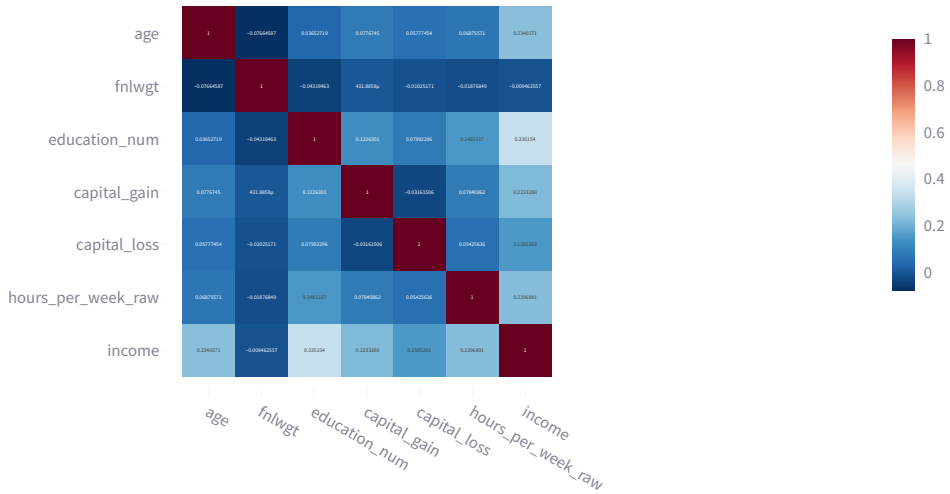
Statistical Summary

	age	fnlwgt	education_num	capital_gain	capital_loss	hours_per_week_raw	income
count	32561	32561	32561	32561	32561	32561	32561
mean	38.5816	189778.3665	10.0807	1077.6488	87.3038	40.4375	0.2408
std	13.6404	105549.9777	2.5727	7385.2921	402.9602	12.3474	0.4276
min	17	12285	1	0	0	1	0
25%	28	117827	9	0	0	40	0
50%	37	178356	10	0	0	40	0
75%	48	237051	12	0	0	45	0
max	90	1484705	16	99999	4356	99	1

Interactive Visualizations

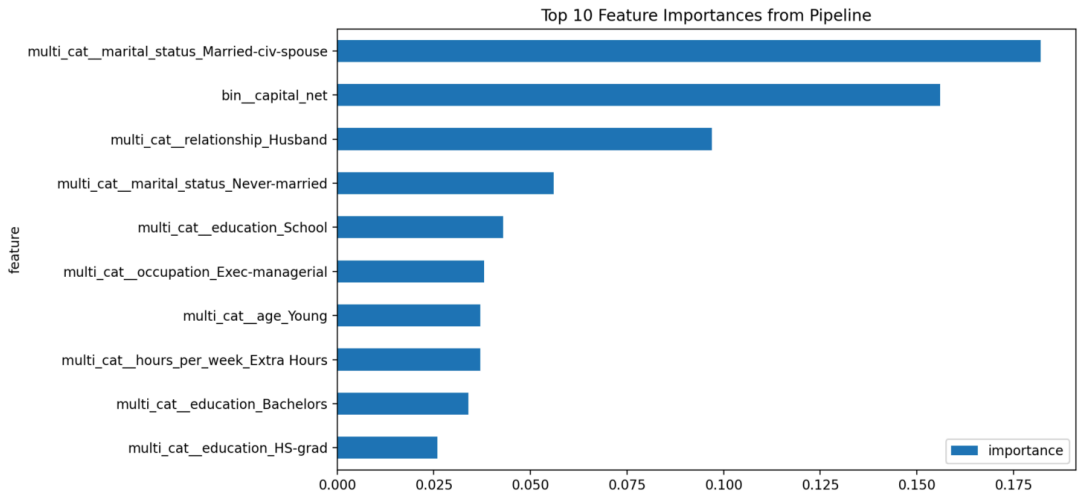
Correlation Distribution Categorical Analysis Target Analysis

Correlation Heatmap



Random Forest Model Training & Evaluation

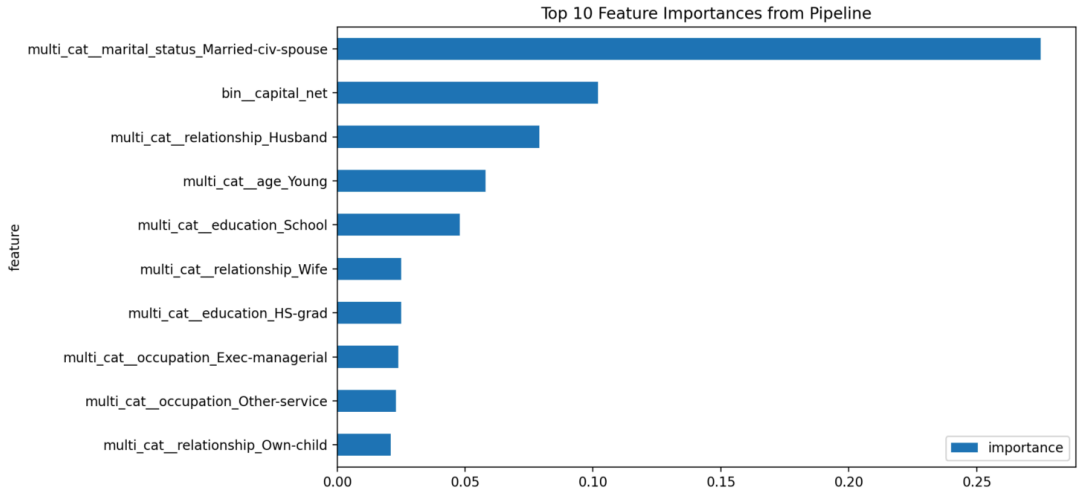
Accuracy	Precision	Recall	F1 Score
0.845	0.700	0.626	0.661



	feature	importance
21	multi_cat__marital_status_Married-civ-spouse	0.182
2	bin__capital_net	0.156
40	multi_cat__relationship_Husband	0.097
23	multi_cat__marital_status_Never-married	0.056
14	multi_cat__education_School	0.043
29	multi_cat__occupation_Exec-managerial	0.038
16	multi_cat__hours_per_week_Extra Hours	0.037
6	multi_cat__age_Young	0.037
9	multi_cat__education_Bachelors	0.034
11	multi_cat__education_HS-grad	0.026

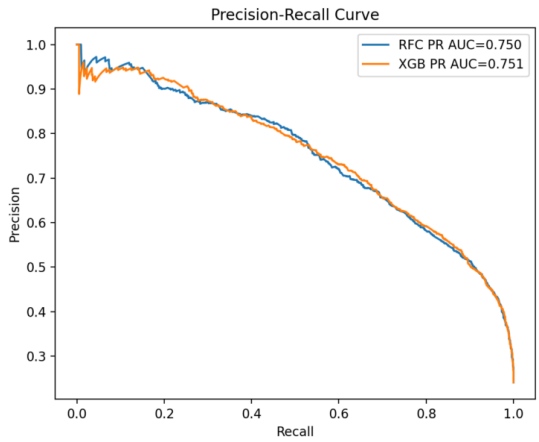
XGBoost Model Training & Evaluation

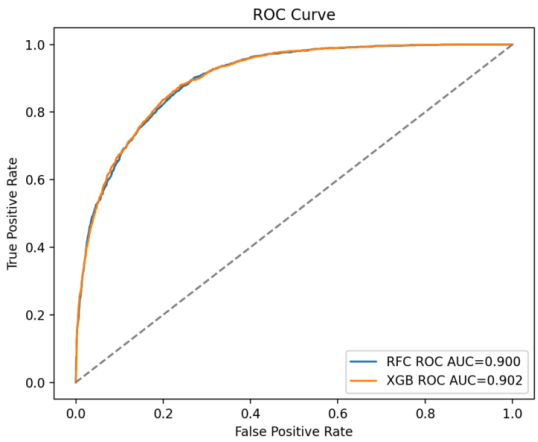
Accuracy	Precision	Recall	F1 Score
0.849	0.705	0.643	0.673



	feature	importance
21	multi_cat__marital_status_Married-civ-spouse	0.275
2	bin__capital_net	0.102
40	multi_cat__relationship_Husband	0.079
6	multi_cat__age_Young	0.058
14	multi_cat__education_School	0.048
11	multi_cat__education_HS-grad	0.025
45	multi_cat__relationship_Wife	0.025
29	multi_cat__occupation_Exec-managerial	0.024
33	multi_cat__occupation_Other-service	0.023
43	multi_cat__relationship_Own-child	0.021

	Model	Test AUC	Test F1 Macro	Test F1 Micro
0	RandomForest	0.9004	0.7805	0.8454
1	XGBoost	0.9017	0.7875	0.8494





Try a Prediction

Edit the sample values below and predict income level.

age: Old	workclass: Private
education: HS-grad	education_num: 9.00
marital_status: Widowed	occupation: Prof-specialty
relationship: Not-in-family	race: White
sex: Male	native_country: Other
capital_net: Major	hours_per_week: Extra Hours

Predict Income

Predicted Income: >50K

Confidence: 89.09%

Dataset Assistant Chatbot

Ask something about the dataset or model...